

Steady state condition:-

Stable probability:- In many markov chains, the probability for a particular state will approach a limiting value as time goes to infinity. In other words, in the far future, the probability wont be changing much from one transition to the next transition. These limiting values are called stable probabilities.

Steady state condition:- If a system is such that each state has the prob. equal to its stable probability, the prob. will persist for all the time. Then the system is said to be in steady state ~~condition~~ condition.

Steady state vector (or) Equilibrium vector of a Markov chain.

If a Markov chain with transition matrix P is regular, then there is a unique vector V such that, for any prob. vector V and for large values of n , $V \cdot P^n = V$. Vector V is called the equilibrium vector or fixed vector or steady state vector of the markov chain.

This is also called ~~as~~ long range trend of the markov chain.

Note:- If a Markov chain with transition probability matrix P is regular, then there exists a probability vector V such that $VP = V$.

prob vector:- A prob. vector is a matrix of only one row having non-negative entries with sum of the entries equal to 1.

prob1: A housewife buys 3 kinds of cereals A, B and C. She never buys the same cereals in successive weeks. If she buys Cereal A, the next week, she buys B. However if she buys B or C, the next week it is 3 times as likely to buy A as other cereals. In the long-run, how often does she buy each of the three cereals.

So The t.p.m. of the purchasing by housewife can be written as

$$P = \begin{matrix} & \begin{matrix} A & B & C \end{matrix} \\ \begin{matrix} A \\ B \\ C \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 3/4 & 0 & 1/4 \\ 3/4 & 1/4 & 0 \end{bmatrix} \end{matrix}$$

If she buys B (or C) the next week it is 3 times as likely to buy A as other cereals. From B, the possible next cereals are A or C (note)

Let $P(B \rightarrow C) = x$, then

$$P(B \rightarrow A) = 3x.$$

$$3x + x = 1$$

$$\Rightarrow x = 1/4$$

$$\text{So, } P(B \rightarrow C) = 1/4, \\ P(A \rightarrow B) = 3/4,$$

Similarly

$$P(C \rightarrow B) = 1/4$$

$$\& P(C \rightarrow A) = 3/4.$$

Let $V = \{v_0, v_1, v_2\}$ be the steady state prob. distribution of the markov chain.

using $VP = V$, we have.

$$[v_0, v_1, v_2] \begin{bmatrix} 0 & 1 & 0 \\ 3/4 & 0 & 1/4 \\ 3/4 & 1/4 & 0 \end{bmatrix} = [v_0, v_1, v_2]$$

$$\Rightarrow \frac{3}{4}v_1 + \frac{3}{4}v_2 = v_0 \quad \text{--- (1)}$$

$$v_0 + \frac{1}{4}v_2 = v_1 \quad \text{--- (2)}$$

$$\frac{1}{4}v_1 = v_2 \Rightarrow v_1 = 4v_2 \quad \text{--- (3)}$$

Also $v_0 + v_1 + v_2 = 1$. (According to defn of prob. vector)

from (2) & (3)

$$v_0 + \frac{1}{4}v_2 = 4v_2.$$

$$\Rightarrow v_0 = \frac{15}{4}v_2.$$

substituting all terms in (4)

$$\frac{15}{4}v_2 + 4v_2 + v_2 = 1$$

$$\Rightarrow \left(\frac{15+16+4}{4} \right) v_2 = 1$$

$$\Rightarrow \boxed{v_2 = \frac{4}{35}}$$

$$v_1 = 4v_2 = \frac{16}{35} \quad \left| \quad v_0 = \frac{15}{4}v_2 = \frac{15}{4} \times \frac{4}{35} = \frac{15}{35}$$